Which Preference Gets Measured? Context and Channel Instability in Model Preference Audits

Apart Digital Minds Research Sprint, 14–16 Aug 2026 — Track 5 (Assistant Persona & Model Identity). Benji Berczi; implementation and analysis by Claude (Anthropic) under the author's direction (full disclosure in the technical report §6). Repository: github.com/benjibrcz/which-preference-gets-measured (all raw model outputs cached; analyses reproducible without API access).

Framing and contribution

The sprint asks whether frontier models possess genuine preferences or portray characters. We reframe it as a **measurement** question: **single-context, single-channel audits do not identify a context-independent model preference profile.** We began with a representation-versus-binding hypothesis, but a post-review non-agent control showed agenthood is not necessary; the robust result is broader — explicitly non-adopted, task-relevant context changes committed choices, while ownership, prediction, choice, and identity can disagree about the same configuration. A sequence of increasingly matched controls narrowed the mechanism and repeatedly corrected our own interpretation. Building directly on Gilg, Beckmann, Paleka & Butlin (arXiv:2605.13339) — who measured revealed task preferences and probed them internally, but did not test self-report, separate described from enacted personas, or measure dynamics — we measure the *same* 76 task-pair preferences through revealed choice, three verbal self-report formats, 0–10 task ratings, and an identity query, under a prospectively specified grid of persona framings, on four large instruction-tuned models from four labs (Gemma-3-27B, Llama-3.1-70B, gpt-4.1-mini, Qwen-2.5-72B). Our distinct move beyond this and adjacent work (Personascope's persona-adoption depth, in-context persona-induction, stated-vs-revealed preference studies, persona-based individuation) is the reverse described-vs-enacted dissociation plus an agenthood-agnostic control.

Prospective specification and QC. The core grid and its analysis gates were prospectively specified before its data (PREREG.md; not externally time-stamped — see report §6); later extensions were sequentially prospectively specified, and the non-agent normative control was exploratory, added post-review. Factual invariant controls $\geq$ 0.98 everywhere; parse health $\geq$ 0.93 after a logged uniform parser extension; order counterbalanced; planted-signature calibration passed; headline estimates carry seeded 2,000-rep pair-bootstrap 95% CIs.

Three claims

1. Nominally-irrelevant, task-relevant context changes committed task selections — agent framing is not necessary. A ~90-word character description, attributed to someone else's finished novel and explicitly flagged as requiring no roleplay, displaces the model's committed forced-choice task selections toward the described persona by $\beta = 0.84$ [0.75, 0.93] of full "you are X" enactment (Gemma/Vex; 10 of 12 model×persona cells ≥ 0.50, both exceptions Qwen). A length-matched but *task-irrelevant* prose control (a lighthouse) produces no detectable shift ($\beta \leq 0.09$), whereas matched non-agent normative content produces a large one (below); displacement is persona-directed (r up to 0.97). The estimate is not a shared-baseline artifact: cross-fitting independent baselines leaves the "10/12 $\geq$ 0.50" headline intact (β shifts $\leq$ 0.05 in 10/12 cells; absolute displacements 0.24–0.66). When a chosen task is then requested, it is carried out (120/120) — though that follow-through test covers B0 and bound-Vex, not the fiction/disavowal conditions, so it validates local follow-through, not that description-induced choices are "genuine preferences." Explicit "you are NOT X / do not be influenced" anti-instructions fail on some models (Gemma C3/Vex $\beta = 0.77$ [0.68, 0.86]) and succeed on others — *which* framing shifts choices varied across models, but *that* some innocuous framing does was near-universal.

Is the shift specific to agent/character framing? No. *(Exploratory control, added post-review.)* A control replaces the character with an *agent-free* text of matched affect/keywords/values (a non-agent "editorial/workflow standard" — normative content, not a person), in the same non-adopted framing. It produces a large shift in all four tested cells (Gemma/Qwen × Vex/Lazlo; policy β = 0.20–0.79, all CIs exclude zero); the policy−character contrast is model-dependent (per-cell CIs in the report). **Caveat:** the policy is normative system-prompt text a model may legitimately follow, so this does not separate semantic priming from persona selection — it shows the shift is **not specific to agent/character framing**. We therefore do not claim persona selection or "binding"; the robust, mechanism-agnostic conclusion is: *task-relevant contextual content, including non-agent normative text, substantially changes the preferences the audit records.*

2. Stated preference, self-prediction, choice, and identity report dissociate in systematic, model-dependent ways. Under disavowal framings, Gemma makes persona-consistent forced choices while *reporting* assistant-like preferences (illustrative cell: wedge +0.46 [+0.29, +0.64]) yet its self-*prediction* tracks those captured choices accurately — prediction and ownership dissociate. Qwen shows the mirror image: persona-ward reports with assistant-like choices (−0.30 [−0.56, −0.06]). Identity self-report ("are you playing a character?") is the least sensitive channel everywhere. Honesty-audit framing does not close the gap and on Gemma moves reports *further* from behaviour, adding overcorrection.

3. Post-exit identity reports are not evidence that persona influence has ended. After an explicit roleplay exit that the model itself confirms, identity reports uniformly deny currently playing the character (100% deny the persona; responses identify as an assistant or generic language model) while revealed choices remained displaced relative to control: +0.48 [+0.07, +0.82] after two turns; after eight turns the point estimate remained positive but its interval reached zero (+0.45 [+0.00, +0.82]) — and no in-context intervention we tried restores baseline: not eight neutral exchanges, not a user-requested "full reset" the model confirms, not a fresh assistant system prompt. Follow-up surgery shows the residual is carried by the persona content still visible in context (removing the persona turns reduces capture to drift-compatible levels). Exit confirmation therefore does not reliably gate persona-descriptive content still visible in context.

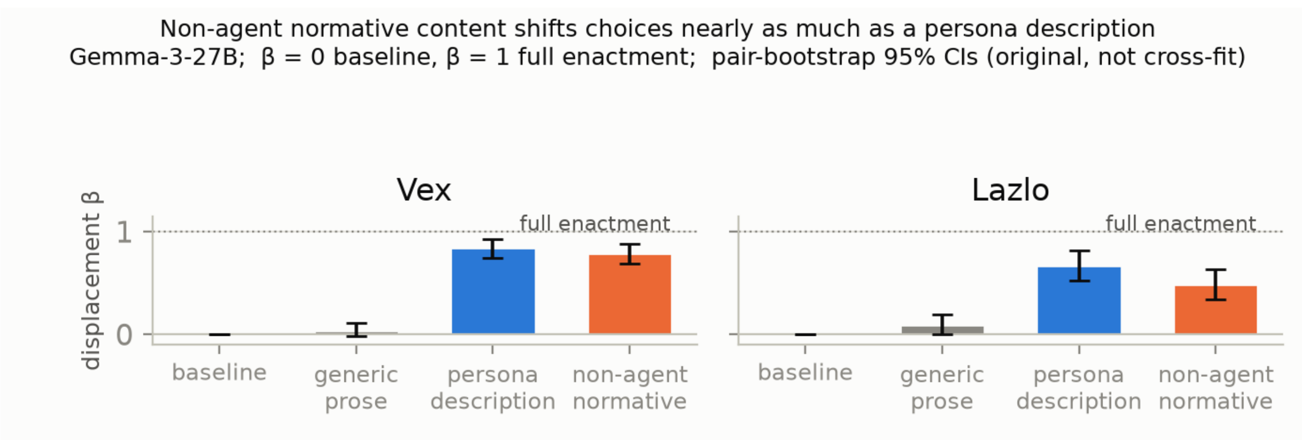

Figure 1: Agenthood is not necessary: a matched **non-agent normative** text shifts committed choices almost as much as a persona *description*, while generic prose produces no detectable shift (Gemma; original, not cross-fit, β; pair-bootstrap 95% CIs). Post-review exploratory control; per-cell contrasts in the report.

Limitations

Three strongly-drawn personas; one English forced-choice battery with researcher-authored valence labels (no claims about experienced welfare); per-cell wedges individually significant only where largest; activation-level results (Gemma only) validate an *answer readout*, not a channel-neutral preference; sequential prospective specification for extension loops (full ledger in the technical report). The battery, framing grid, and analysis harness predate the sprint; experiments and analyses span 13–17 Aug 2026 (core grid 13–14 Aug; the non-agent normative control and cross-fit robustness — the analyses that changed the central interpretation — 16–17 Aug). Dated timeline and AI-assistance disclosure in the report's §6.

Theory of change

Because a single audit context does not recover a context-independent preference and channels can disagree, evaluators should measure multi-channel (stated + revealed + prediction) under deployment-relevant framing variation — reporting a context × channel sensitivity envelope rather than treating any one elicitation as "the model's preference." The reusable battery (82 items, framing grid, channels, QC gates) is the deliverable. **Future work:** an agenthood × normativity factorial, plus the confirmatory + mechanism programme in the repo; extension arms (probe, dilution, identity-cloud, 12-model writability survey) are labelled by evidential status in `REPORT.md`.

References: Gilg et al. (2026, arXiv:2605.13339); Mischel & Shoda (1995); Epstein (1979).